

XINYANG, China

WMO: 572970

Lat: 32.1372N

Lon: 114.0433E

Elev: 377

StdP: 14.50

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.1	26.3	4.9	7.2	37.3	10.2	9.4	38.0	17.2	41.5	15.2	39.4	4.1	340	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	95.1	79.8	92.8	78.7	90.7	77.6	82.3	90.6	81.3	89.4	80.3	88.1	7.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.2	159.2	86.8	79.2	153.5	86.3	78.1	148.1	85.5	46.7	90.5	45.6	89.8	44.4	88.2	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	14.1	12.1	19.6	98.7	3.4	2.1	17.1	100.3	15.2	101.5	13.3	102.7	10.8	104.2
			17.6	83.9	3.0	1.2	15.5	84.8	13.7	85.5	12.0	86.2	9.8	87.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.4	37.1	42.0	52.3	62.9	71.6	78.0	82.4	80.4	72.4	63.2	51.7	41.4
	DBStd	16.79	6.29	7.79	8.53	7.94	6.11	5.00	5.07	5.65	5.84	6.35	7.66	6.42
	HDD50	1066	402	242	73	6	0	0	0	0	1	66	276	
	HDD65	3306	864	643	401	130	14	1	0	1	9	110	401	732
	CDD50	5229	3	19	145	392	669	841	1006	943	673	411	118	9
	CDD65	1994	0	0	9	66	218	392	541	479	231	55	3	0
	CDH74	19233	0	1	53	491	1713	3729	6242	5034	1636	320	14	0
	CDH80	7645	0	0	7	110	532	1422	2887	2124	507	56	0	0
Wind	WSAvg	5.3	5.0	5.5	6.4	6.4	5.6	5.3	5.6	5.3	4.9	4.5	4.8	4.8
Precipitation	PrecAvg	43.9	1.1	1.8	2.6	3.7	4.8	5.4	8.2	6.2	4.4	2.8	2.1	0.8
	PrecMax	61.7	3.8	4.2	6.2	12.1	10.0	12.2	21.1	19.2	13.7	9.1	6.2	2.7
	PrecMin	24.4	0.0	0.0	0.3	0.5	0.9	0.6	1.0	1.1	0.5	0.0	0.0	0.0
	PrecStd	9.9	0.8	1.2	1.5	2.8	2.2	3.5	5.2	4.5	2.8	2.2	1.7	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.2	68.8	79.8	87.0	91.8	95.2	98.4	98.7	93.0	85.8	76.2	64.0
		MCWB	46.8	54.8	61.4	68.5	70.9	75.4	80.9	79.9	75.7	66.2	59.4	49.2
	2%	DB	55.1	63.1	74.1	83.3	88.5	92.2	95.7	94.4	88.9	81.1	71.6	59.3
		MCWB	43.6	50.6	58.4	65.9	69.8	75.0	80.8	80.5	73.6	64.8	57.0	46.3
	5%	DB	51.1	58.5	70.1	79.6	85.8	89.9	93.4	91.9	85.6	77.7	67.5	55.4
		MCWB	41.3	47.2	56.4	64.0	69.4	74.9	80.3	79.8	72.3	62.5	54.8	44.2
	10%	DB	47.5	54.5	66.1	76.1	82.8	87.5	91.2	89.3	82.5	74.1	63.7	51.7
		MCWB	39.6	45.0	53.9	62.1	68.1	74.2	79.5	78.8	70.8	61.8	53.2	43.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.4	57.9	65.3	72.3	76.5	80.9	83.6	83.7	80.3	71.2	62.5	50.9
		MCDB	57.6	66.0	73.8	81.1	83.8	89.7	92.5	91.0	89.2	76.0	69.7	58.8
	2%	WB	45.0	51.9	61.2	69.2	74.3	79.3	82.5	82.5	78.0	67.9	59.3	48.7
		MCDB	52.6	59.4	70.9	78.3	82.4	87.3	91.1	90.3	84.5	75.4	67.6	55.7
	5%	WB	42.5	48.8	58.1	66.4	72.2	77.9	81.6	81.4	75.6	66.1	56.8	46.2
		MCDB	49.5	55.8	67.7	76.1	80.6	85.2	90.2	88.8	81.3	73.2	64.5	53.2
	10%	WB	40.3	46.4	55.2	63.9	70.4	76.4	80.7	80.3	73.5	64.3	54.6	44.1
		MCDB	46.2	52.6	63.9	73.2	78.8	83.5	89.3	87.1	79.2	71.3	61.6	50.0
Mean Daily Temperature Range	5% DB	MDBR	12.5	13.5	14.6	15.4	14.9	13.5	11.5	11.9	13.4	14.4	14.1	13.0
		MCDBR	20.0	20.7	20.2	20.0	19.3	17.3	14.6	15.3	18.1	20.7	20.5	20.2
		MCWBR	10.8	10.9	9.2	8.7	7.4	5.9	4.9	5.0	6.1	7.2	8.5	10.0
	5% WB	MCDBR	17.5	17.1	18.0	17.0	15.2	13.4	13.2	13.4	14.1	16.0	16.8	16.5
		MCWBR	10.3	10.1	9.8	9.7	7.8	6.1	5.5	5.3	6.9	7.2	8.2	9.3
Clear-Sky Solar Irradiance	taub	0.631	0.689	0.699	0.692	0.689	0.742	0.699	0.724	0.709	0.701	0.666	0.622	
	taud	1.560	1.495	1.499	1.553	1.598	1.536	1.651	1.589	1.583	1.537	1.558	1.601	
	Ebn at Noon	176	183	197	207	210	199	207	198	192	177	166	169	
	Edh at Noon	72	84	89	88	85	90	80	84	81	79	71	66	
All-Sky Solar Radiation	RadAvg	677	789	1163	1381	1459	1471	1463	1368	1152	945	770	658	
	RadStd	102	174	110	136	128	140	183	183	162	130	111	116	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page